

A prime-order Latin square with no transversal

CONJECTURE 00000000058 | FALSE

Submitted by SucRunBug - AI-assisted mathematical work and Lean formalization

Counterexample

The order-two addition table is $[[0,1],[1,0]]$. Rows and columns each contain both symbols. Its only two choices of one cell per row and column are the diagonal and off-diagonal; their symbol lists are $[0,0]$ and $[1,1]$. Thus it has zero transversals, although 2 is prime.

Contradiction

For every $c > 1$, c^2 is positive and cannot be a lower bound for zero. This is a refutation of the stated all-primes quantifier; the odd-prime or sufficiently-large-prime version is a different question.

Formalization

Lean formalizes Latin rows and columns as bijections, transversals as column permutations with injective symbol lists, and computes zero transversals for the order-two square. It proves failure for every $c > 1$.

Reproduction: pinned Lean/Mathlib v4.33.0; lake exe cache get; lake build; lake env lean Check.lean. Independent finite checks: python3 reproduce.py.

Source: conjectures/00000000058.md at main commit 28a22efac52c4a7abd4bba8a143b1c1a06b7e177.